Sulfur precursors for MoS2 synthesis

"Which sulfur precursors are used for MoS2 hydrothermal growth?" *Generated by LeapSpace*

MoS_2 hydrothermal growth in the retrieved evidence uses **thiourea**, **L-cysteine**, **L-cystine**, **thioacetamide**, **elemental sulfur / sulfur powder**, and a few **single-source or sulfur-rich precursors** such as **thiomolybdate-based compounds** and **$(NH_4)_2MoS_4$-derived systems**. [132] [133] [134] [135] [136] [137] [138] [139] [140] [141] [142] [143] [144] [145] [146] [147]

Most used sulfur precursors

Sulfur precursor	How it is used in hydrothermal MoS_2 growth	What the evidence suggests
Thiourea	Common sulfur source with sodium molybdate / ammonium molybdate in many hydrothermal syntheses. [135] [144] [146] [147] [148] [149] [150] [151] [152] [153] [154] [155]	One of the **most versatile and frequently used** sulfur precursors; hydrolysis generates H_2S that drives sulfidation. [138] [149] [150]
L-cysteine	Used with sodium molybdate or other Mo sources for hydrothermal MoS_2. [132] [133] [156] [157]	Acts as **both sulfur source and reducing/directing agent** in some systems. [133] [156] [157]
Thioacetamide	Used with sodium molybdate in hydrothermal MoS_2 nanopowder and film growth. [137] [142] [158] [159]	A common sulfur donor; its hydrolysis supplies sulfur species for MoS_2 formation. [142] [149] [150]
Elemental sulfur / sulfur powder	Used as the sulfur precursor in hydrothermal or sulfurization-assisted growth. [134] [139] [160] [161]	Supports MoS_2 formation, but the morphology depends strongly on sulfur availability and growth conditions. [139] [161]
L-cystine	Compared directly with thiourea and sulfur powder in hydrothermal MoS_2 catalyst synthesis. [134]	Another valid organic sulfur source, with morphology and activity depending on precursor choice. [134]
Thiomolybdate / single-source sulfur-containing precursors	Includes ammonium thiomolybdate-related systems and other sulfur-rich precursors that decompose to MoS_2. [162] [163] [164] [165]	Useful when a **single-source** or sulfur-rich route is desired. [162] [165] [166]

Practical takeaways

- If you want the **most standard hydrothermal sulfur source**, **thiourea** is the clearest choice from the evidence. [134] [135] [146] [147] [149] [150]

- If you want a precursor that can also help **direct morphology**, **L-cysteine** is especially useful. [133] [156] [157]
- If you want to tune **nucleation and morphology**, **thioacetamide** and **elemental sulfur** are also well supported, with the sulfur source affecting crystallinity, density, and final shape. [137] [139] [141] [161]

Bottom line: the retrieved literature supports **thiourea, L-cysteine, thioacetamide, elemental sulfur/sulfur powder, L-cystine, and sulfur-rich single-source precursors** as the main sulfur precursors for hydrothermal MoS_2 growth, with **thiourea** appearing most broadly used and **L-cysteine** often acting as both sulfur source and structure-directing agent. [132] [133] [134] [135] [137] [141] [142] [149] [150] [156]

Confidence level: **High** — multiple independent hydrothermal studies directly name these sulfur precursors and repeatedly show that precursor identity affects MoS_2 morphology and growth behavior.

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